

Marcos Galván López

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Professional Profile

AI Engineer with a rare combination of applied mathematics, probabilistic modeling, and hands-on generative AI systems engineering. Passionate about pushing AI toward edge applications, spanning autonomous multi-agent research systems and gradient-free optimization of model weights. Research background in mechanistic interpretability of LLM weight space and fine-tuning techniques, complements hands-on experience architecting and shipping full-stack agentic systems (LangGraph orchestration, multi-model routing, sandboxed execution, and persistent state).

Technical Skills

Programming Languages: Python, JavaScript, C++, C#, Java

Cloud & Infrastructure: AWS, Git/GitHub, Supabase/Postgres, relational databases

Agent & GenAI Engineering: LangGraph agent orchestration, Anthropic Claude SDK and Gemini API integration, multi-model cost-optimization strategies (free-tier vs. on-demand routing), LLM evaluation pipelines, LoRA and zero-order fine-tuning techniques, mechanistic interpretability of weight space, agent-data-structure design

ML & Math Foundations: Monte Carlo methods (MCMC), stochastic calculus

Other: Technical writing and documentation, cross-institutional collaboration, English C1 (TOEFL-iBT), continuous tracking of emerging AI research and tooling

Professional Experience

Banco de México

Mexico City, MX

Informatics Technician

July 2023 – January 2024

- Monitored and maintained database processes supporting critical financial services infrastructure, ensuring data integrity and availability for downstream systems.
- Designed and implemented secure file-transfer protocols between external clients and the central bank's internal systems.

Selected Projects

LoRA-RandOpt (In progress) – LoRA-restricted random search for LLM specialization *GitHub*

- Built and documented an evaluation pipeline for Qwen2.5-1.5B-Instruct on BBH Boolean Expressions benchmarks under GPU-constrained hardware.
- Implemented zero-shot baselining, top-K majority-vote ensembling, and rigorous statistical comparison (Mann–Whitney U) between fine-tuning strategies.

Education

Centro de Investigación en Computación (CIC)

Mexico City, MX

Ongoing – MSc in Computer Science. GPA 3.9

Graduation Date: 01/27

- Research in probabilistic AI for semantically accurate latent-space representation and its application to pattern classification and zero-shot learning.

University of Edinburgh (UoE)

Edinburgh, UK

Academic Visit – Artificial Intelligence Applications Institute

September 2025 – December 2025

- Collaborated with Dr. Vaishak Belle on a probabilistic Neuro-Symbolic framework for shortcut-reasoning mitigation.
- Research resulted in co-authorship of a paper accepted at NeSy 2026 (Lisbon, Portugal).

Instituto Politécnico Nacional (IPN)

Mathematical Engineer. GPA 3.6

Mexico City, MX

Graduation Date: 08/23

Thesis: Comparative study of control variables for variance reduction in financial asset valuation via stochastic calculus and Monte Carlo simulation.

Publications & Talks

- Galván-López, M., Upreti, N., & Belle, V. *Soft Symbol Grounding for Prototypical Concepts* (Accepted). 20th International Conference on Neurosymbolic Learning and Reasoning (NeSy 2026), Lisbon, Portugal, April 2026.
- Galván-López, M., Calvo, H., & Valdez-Rodríguez, J. E. (2025). *Representation of Neuro-Symbolic Networks for Sub-algebraic Terms*. Mexican Congress on Artificial Intelligence, pp. 174–186. Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-97910-1_14

Extra-Curricular & Academic Service

- **Judge & Speaker**, AI Safety Hackathon by Apart Research
- **Mentor**, Apart Research AI Safety Fellowship
- **Creator & Instructor**, "Mi Primer GPT" – introductory course on language generation with Transformers; first course of its kind delivered in Spanish
- Member, Mexican Association of Artificial Intelligence (SMIA)